

# Markups and Markdowns in Indian Organised Manufacturing: Plant-Level Evidence from the Annual Survey of Industries, 2012–2022

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## Abstract

We provide the first joint plant-level measurement of product-market power (markups) and labour-market power (wage markdowns and monopsony) for modern Indian organised manufacturing, using 423,127 plant-year observations on 142,493 factories from the Annual Survey of Industries, 2012–13 to 2022–23. Markups are estimated with the production approach of [De Loecker & Warzynski \(2012\)](#) under a translog gross-output technology with plant-specific elasticities, identified by the [Akerberg et al. \(2015\)](#) control-function moments; markdowns follow the two-flexible-input wedge of [Brooks et al. \(2021\)](#). Four facts emerge. First, product-market power is modest and stable: the median markup sits in a narrow 1.09–1.74 band across industries, around 1.22 in the aggregate, declining gently after 2015, while employment concentration fell throughout (mean industry HHI 0.086 to 0.066). Second, labour-market power is pervasive: 41–44 per cent of plants pay labour below its marginal revenue product, implying a typical-plant labour-supply elasticity near two; across three aggregation concepts the implied elasticity spans 0.3–2. Third, the two margins have opposite geographies: markups are spatially uniform across the industrial core, whereas monopsony incidence ranges from 0.09 to 0.63 across states and is systematically higher in rural plants (44.2 versus 38.4 per cent urban, in every sample year, with gaps above twenty percentage points in West Bengal

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and Uttar Pradesh). Fourth, measurement discipline strengthens the asymmetry: under factor-augmenting technology the markup level falls by ten per cent while labour elasticities rise by a third, so measured product-market power is an upper bound and labour-market power a lower bound. The static welfare cost of markup dispersion (5.8–9.9 per cent of sales) exceeds the monopsony triangle (2.2–2.8 per cent of the wage bill), but both margins matter most through the misallocation they induce, quantified in the companion paper.

**JEL:** D22, D24, D43, J42, L11, L13, L60, O14. **Keywords:** markups; markdowns; monopsony; production function; ACF; translog; ASI; India.

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# 1 Introduction

Market power has two faces. A firm with power in its *product* market prices above marginal cost — the markup  $\mu \equiv p/c > 1$  — transferring surplus from consumers and, when markups differ across producers, distorting the allocation of demand. A firm with power in its *labour* market pays a wage below the marginal revenue product of labour — the markdown  $\psi \equiv \text{MRPL}/w > 1$  — transferring surplus from workers and distorting the allocation of labour. The two faces are formally symmetric (Robinson, 1933): monopoly restricts output to raise price; monopsony restricts hiring to lower the wage. The static distortion each wedge implies is textbook but worth stating. A monopolist facing demand elasticity  $\eta$  restricts output until  $p = \frac{\eta}{\eta-1}c$ ; a monopsonist facing labour-supply elasticity  $\varepsilon_L$  restricts employment until

$$w = \frac{\varepsilon_L}{1 + \varepsilon_L} \text{MRPL} \quad \implies \quad \frac{L}{L^*} = \left( \frac{\varepsilon_L}{1 + \varepsilon_L} \right)^{\varepsilon_L} < 1, \quad (1)$$

so an elasticity of two — our typical-plant estimate — implies employment eleven per cent below its competitive level at the affected plants, before any cross-plant misallocation is counted. They are also, in production data, inseparable: as Section 4 makes precise, the object the production approach recovers from any single input is the *product* of the two wedges, so credible measurement requires estimating both at once. That joint measurement, for the universe of India’s organised manufacturing plants over 2012–2022, is this paper’s task.

The question matters for India with particular force, for three reasons grounded in theory and in the country’s policy debate. First, the welfare economics of the two wedges differ. A common markup is a transfer with modest Harberger losses; *dispersed* markups misallocate inputs across plants and reduce aggregate productivity (Baqae & Farhi, 2020; Edmond et al., 2023) — and India is the canonical high-dispersion economy (Hsieh & Klenow, 2009). Whether Indian markups are large, and above all whether they are dispersed, is thus an input into the country’s productivity accounting, which our companion paper takes up. Second, the labour-market face bears on India’s central development challenge: manufacturing wages and the labour share. If plants systematically pay below marginal product, the gains from industrial growth accrue disproportionately to profits, and policies that raise labour demand or worker mobility carry first-order efficiency gains, not merely distributional ones (Manning, 2003; Berger et al., 2022). Third, the global context sharpened the stakes: the documented rise of markups in the United States (De Loecker et al., 2020) and the finding that measured *markdowns*, not markups, drive it (Yeh et al., 2022) raise the question of where a large developing manufacturing sector sits on both margins.

Stated as questions, the paper asks three things. **Q1 (product market):** how large

are markups in Indian organised manufacturing, how dispersed are they across plants, and how have they moved over 2012–2022 — and through which margin (within-plant pricing, or reallocation of sales across plants)? **Q2 (labour market):** what fraction of plants pay labour below its marginal revenue product, how large is the implied wedge, and what labour-supply elasticity does it reveal workers to face? **Q3 (geography):** do the two wedges share a geography — across states, and across the rural–urban boundary within states — or do product-market and labour-market power live in different places? The identifying lever for all three is the same: cost-minimisation ties output elasticities to revenue shares input by input, so a flexible input priced without frictions (materials) reveals the output wedge, and the *gap* between the labour-based and materials-based versions of the same ratio reveals the labour wedge. Everything else in the paper is the careful implementation of that lever.

Answering these questions faces four well-known challenges, which discipline our design. **(i) Elasticity identification.** The markup formula  $\mu = \theta^X / \alpha^X$  (output elasticity of a flexible input over its revenue share) is only as good as the output elasticity, and input choices respond to unobserved productivity — the classic simultaneity problem (Marschak & Andrews, 1944). We estimate gross-output production functions separately for each 2-digit industry with the control-function approach (Olley & Pakes, 1996; Levinsohn & Petrin, 2003), implementing the Akerberg et al. (2015) correction, whose two-stage moment structure we spell out in equations (5)–(7). **(ii) Technology restrictiveness.** Cobb–Douglas forces one elasticity per industry, assuming away the plant heterogeneity in market power that is the object of interest; our headline technology is therefore the translog, with plant-specific elasticities (4), estimated by a manual two-step GMM. **(iii) Revenue data.** Like nearly all plant surveys, the ASI records revenues, not quantities; the estimated objects are revenue elasticities, and markup *levels* inherit the caveats of Bond et al. (2021) and Raval (2023). We respond in three ways: our questions emphasise dispersion, trends, incidence, and geography, which are robust to level scalings; we cross-validate two estimators and report where they disagree; and we bound the Hicks-neutrality bias directly with a factor-augmenting re-estimation (Section 6) following Demireu (2025). **(iv) The markdown’s identification.** The markdown is the *ratio* of two elasticity-to-share ratios (8); it requires both labour and materials to be flexible and is sensitive to reporting anomalies in factor shares. We document two industries (printing; other transport equipment) where value-added implied by reported shares is negative for a seventh to a third of plants, and exclude them from markdown aggregates on stated, testable grounds.

We bring these methods to a newly constructed panel. From the raw block-level ASI files with common factory identifiers we build a factory-year panel of 935,975 observations (2008–2022), deflate to 2011-12 prices with industry-matched WPI series, and retain for estimation the 423,127 plant-years (142,493 factories, 23 industries) that are operating,

have positive real output, materials, labour and capital, and appear in at least two consecutive rounds — the panel requirement of the ACF moments. Appendix A documents the construction; every number in the paper descends from logged, re-runnable scripts.

Four sets of findings emerge. *First, modest and stable product-market power.* The plant-weighted median markup lies near 1.22, drifting down after 2015; the sales-weighted aggregate is flat at 1.16–1.21, and a Melitz & Polanec (2015) decomposition attributes the post-2016 stability to within-plant compression offset by reallocation toward high-markup plants. Concentration corroborates: the mean industry employment HHI fell from 0.086 to 0.066 and the four-firm ratio from 0.39 to 0.32. *Second, pervasive monopsony.* The markdown exceeds one for 41–44 per cent of plants (36–45 per cent year by year), with a typical-plant implied labour-supply elasticity  $\varepsilon_L = 1/(\psi - 1) \approx 2$ , in the range of international estimates; across three aggregation concepts — typical-plant, employment-weighted, sales-weighted — the implied elasticity spans 0.3 to 2, a sensitivity we document rather than hide. *Third, opposite geographies.* Median markups vary by barely 0.06 across India’s industrial core states, but monopsony incidence spans 0.09–0.63 across states, peaks in the central-eastern mineral belt, and is higher among rural plants than urban in *every* year (44.2 versus 38.4 per cent pooled), with rural–urban gaps above twenty points in West Bengal and Uttar Pradesh and reversals only where rural manufacturing is plantation-scale (Bihar, Assam). Figures 1–2 map both facts. *Fourth, measurement makes the contrast conservative.* Re-estimating the technology with labour-augmenting heterogeneity lowers the median markup to 1.118 (a ten per cent overstatement under Hicks-neutrality, significant at two bootstrap standard errors in 18 of 23 industries) and raises labour elasticities by roughly a third — so the paper’s central asymmetry, small markups against large markdowns, is a lower bound.

The paper contributes to three literatures. To the *markup measurement* literature it adds a large, modern, developing-country panel estimated under the field’s preferred technology with full cross-estimator diagnostics, and one of the first applications bounding the Hicks-neutrality bias on the same plants. To the *monopsony* literature it adds the first economy-wide markdown geography for India — state and rural/urban — and the demonstration that the aggregation concept, not the estimation, drives the wide range of published labour-supply elasticities. To the *Indian industrial development* literature it revises the received picture: after two decades in which the 1990s liberalisation dominated the evidence (De Loecker et al., 2016), we show that in the 2010s–20s the binding margin of market power in Indian factories is not the price of goods but the price of labour.

Table 1 collects the headline numbers for reference throughout.

Section 2 situates the paper. Section 3 describes the institution, the panel, and every variable. Section 4 develops the framework equation by equation. Section 5 presents results; Section 6 the robustness and measurement analysis; Section 7 the welfare accounting; Section 8 concludes. Appendices cover construction (A), industry-level estimates (B),

**Table 1:** Findings at a glance (pooled 2012–2022 unless noted).

| Object                                    | Estimate                                              |
|-------------------------------------------|-------------------------------------------------------|
| Estimation sample                         | 423,127 plant-years; 142,493 factories; 23 industries |
| Median markup $\mu$ (translog, aggregate) | $\approx 1.22$ ; industry medians 1.09–1.74           |
| Sales-weighted aggregate markup           | 1.16–1.21, flat                                       |
| Monopsony incidence $\Pr(\psi > 1)$       | 41–44% pooled; 36–45% by year, declining              |
| Typical-plant implied $\varepsilon_L$     | $\approx 2$ ; concept range [0.3, 2]                  |
| Employment HHI (NIC-4 mean)               | 0.086 (2012) $\rightarrow$ 0.066 (2022)               |
| Rural vs urban monopsony incidence        | 44.2% vs 38.4%, every year                            |
| Largest state rural–urban gaps            | West Bengal +20.8pp; Uttar Pradesh +20.5pp            |
| Factor-augmenting median markup           | 1.118 (vs 1.240 translog)                             |
| Product DWL / labour DWL                  | 5.8–9.9% of sales / 2.2–2.8% of wage bill             |

the Cobb–Douglas benchmark (C), and the aggregation-concept algebra (G).

## 2 Related literature

**The production approach to markups.** Hall (1988) observed that under imperfect competition the Solow residual co-moves with input growth scaled by the markup; De Loecker & Warzynski (2012) turned this into a firm-level estimator: for any input free of adjustment frictions, cost minimisation equates the markup to the ratio of that input’s output elasticity to its revenue share. The approach requires no demand-side data and travels across industries, which made it the workhorse of the market-power revival (De Loecker et al., 2020). Its identification content, however, is only as strong as the production-function estimate behind the elasticity, and the subsequent debate — revenue versus quantity elasticities (Bond et al., 2021), the input-choice internal-consistency test (Raval, 2023), non-parametric identification (Gandhi et al., 2020), and factor-augmenting technology (Demirer, 2025; Rubens et al., 2025) — shapes our estimation and robustness choices throughout.

**Monopsony and markdowns.** The theory descends from Robinson (1933): an upward-sloping labour-supply curve to the firm implies  $w = \text{MRPL}/(1 + 1/\varepsilon_L)$ , so the wedge  $\psi = 1 + 1/\varepsilon_L$  maps one-for-one into the firm-level labour-supply elasticity. Modern work grounds the slope in search frictions and idiosyncratic preferences (Manning, 2003; Card et al., 2018) and embeds it in general equilibrium with granular employers (Berger et al., 2022). Measurement proceeds by direct elasticity estimation, by concentration proxies (Azar et al., 2022), or — our route — by the production approach with two flexible inputs (Brooks et al., 2021; Yeh et al., 2022; Morlacco, 2020), which delivers a plant-level  $\psi$  from the same estimation that delivers  $\mu$ . For developing countries, Brooks et al. (2021) document markdown-driven labour shares in India and China, and Amodio & de Roux (2023) substantial plant wage-setting power in Colombia; economy-wide Indian evidence at the plant level, and any evidence on its geography, has been absent.

**Magnitudes elsewhere.** For calibration of what “large” means: De Loecker et al. (2020) report US sales-weighted aggregate markups rising from 1.21 (1980) to 1.61 (2016), driven by the upper tail and by reallocation; subsequent scrutiny attributes part of the level and trend to elasticity measurement, cost-share construction, and financial-statement data (Bond et al., 2021; Raval, 2023). On the labour side, Yeh et al. (2022) estimate an average US markdown of 1.53 — workers paid 65 cents per marginal dollar — and, strikingly, that reassessed *markdowns* rather than markups carry the secular “market power” rise. Our Indian numbers should be read against these: an aggregate markup an order of magnitude flatter and lower than the US series, and a monopsony incidence entirely comparable to rich-country estimates.

**Indian labour-market institutions.** The regulation literature provides the institutional backdrop for our geography: state-level employment protection shapes manufacturing outcomes (Besley & Burgess, 2004), size-contingent statutes (Industrial Disputes Act Chapter V-B at 100 workers; EPF/ESI thresholds near 20) create discontinuities in effective labour costs (Bertrand et al., 2021), and the organised sector’s small employment footprint keeps outside options thin outside metropolitan labour markets — the thinness our rural–urban results measure directly.

**India.** The 1990s liberalisation generated the classic Indian market-power evidence: pro-competitive discipline (Krishna & Mitra, 1998), productivity gains through output- and input-tariff channels (Topalova & Khandelwal, 2011), and the landmark price-decomposition of De Loecker et al. (2016) showing markups *rising* with liberalisation as input costs fell faster than prices. Sivadasan (2009) document level effects of reform on productivity. What the Indian literature lacks is precisely what panels since 2010 permit: joint plant-level markups and markdowns on the post-reform economy. That gap — and its spatial dimension, invisible in the industry-level studies — is the one this paper fills; the determinants of the objects measured here, including the 2018+ protection reversal, are analysed in the companion determinants paper.

### 3 Institutions, data, and variables

#### 3.1 The Annual Survey of Industries

The ASI is the census-cum-sample survey of India’s registered factories (Factories Act 1948: ten or more workers with power, twenty without), conducted under the Collection of Statistics Act. Larger units (including all with 100+ employees) are enumerated every year (the census scheme); the remainder are sampled with known probabilities, and inverse-probability multipliers  $w_{it}$  (‘wt’) are supplied. The unit of observation is the factory (establishment). We use the unit-level distribution with *common factory identifiers* — an eight-character code, constant for a plant across years, whose anatomy we verified

against the official record layouts: characters 1–5 are a serial, characters 6–7 the 2011-census *state* code (empirically confirmed: frequencies peak at Tamil Nadu, Maharashtra, Gujarat, Uttar Pradesh in that order), and character 8 a scheme character. Block A records each factory’s rural/urban sector (item A9), which our geography exploits.

### 3.2 Panel construction in brief

Each round is distributed as ten fixed-width block files (A–J) whose field positions vary across six format eras; all were parsed against a column dictionary built from the official layout records. Multi-item blocks (fixed assets; employment; indigenous and imported materials; products) are collapsed to factory totals, blocks merged one-to-one on the identifier, and rounds appended: 935,975 factory-years, 2008–2022, with factory×year verified unique. Operating-unit filters (positive output, materials, labour, capital, multiplier) and the 2011-12-base deflators (available from April 2012) yield the 2012–2022 real panel of 495,148 plant-years; the ACF moment’s requirement of a lagged observation leaves the estimation panel of **423,127 plant-years on 142,493 factories** in manufacturing divisions NIC 10–33. Appendix A gives the full chain and validation.

### 3.3 Variable definitions

Table 2 defines every variable used. Two conventions deserve emphasis. Revenue shares are *nominal* —  $\alpha^M = p^M M / (pY)$  and  $\alpha^L = wL / (pY)$  with both numerator and denominator at current prices — because the theory’s shares are expenditure shares and deflators cancel; production-function inputs and output are *real*, deflated at 2011-12 base with industry-matched WPI for output and materials (petroleum and repair fallbacks documented), the machinery WPI for capital, and labour entering as a headcount.

### 3.4 The macro backdrop, 2012–2022

The decade supplies unusual variation against which to read trends. Trade openness fell from 55.8 per cent of GDP (2012) to 37.8 (2020) before recovering to 50.7; applied manufacturing tariffs turned upward after 2018 after two decades of decline; the Goods and Services Tax (2017) and the Insolvency and Bankruptcy Code (2016) reshaped formalisation and exit; and the pandemic delivered a −5.8 per cent output year followed by a +9.7 per cent rebound. Where these forces *cause* movements in market power is the subject of the companion determinants paper; here they caution against reading any single year, which is why our statements are decade-long and our welfare series annual.

**Table 2:** Variable definitions

| Symbol                         | Definition               | ASI source (blocks)                                                     |
|--------------------------------|--------------------------|-------------------------------------------------------------------------|
| $Y_{it}$                       | Real gross output        | Ex-factory value of products (J), deflated by NIC-2 WPI                 |
| $M_{it}$                       | Real materials           | Indigenous (H) + imported (I) inputs, deflated by NIC-2 WPI             |
| $L_{it}$                       | Labour                   | Total persons engaged (E)                                               |
| $K_{it}$                       | Real capital             | Net fixed assets (C), deflated by machinery WPI                         |
| $wL_{it}$                      | Wage bill (nominal)      | Wages + bonus + PF + welfare (E; A for 2016+ formats)                   |
| $p^M M_{it}$                   | Materials bill (nominal) | Purchase value totals (H+I)                                             |
| $\alpha_{it}^M, \alpha_{it}^L$ | Nominal revenue shares   | Materials (wage) bill over nominal output                               |
| $w_{it}$ (‘wt’)                | Survey multiplier        | Frame (A); stored always, applied in aggregation only                   |
| $state_{it}$                   | 2011-census state code   | Characters 6–7 of the factory identifier (A)                            |
| $rural_{it}$                   | Rural (1) / urban (2)    | Item A9 (A), all rounds                                                 |
| NIC2/3/4/5                     | Industry (NIC-2008)      | Activity code (A); no cross-classification concordance needed in-window |

### 3.5 Cost shares at a glance

The two revenue shares that enter every object in the paper are heavily dispersed and heavy-tailed (statistics on the manufacturing estimation sample): the materials share  $\alpha^M$  has median 0.756 with a 10th–90th percentile span of 0.462–0.959, and the labour share  $\alpha^L$  median 0.098 with span 0.023–0.347. Two features matter downstream: the dominance of materials (which stabilises markup measurement, Section 4.1) and a pathological upper tail — before cleaning, 69,031 observations report  $\alpha^L + \alpha^M > 1$ , two-thirds of them concentrated in the two industries flagged in Section 4.5.

**Table 3:** Nominal revenue shares, estimation sample.

|                            | p10   | median | p90   |
|----------------------------|-------|--------|-------|
| Materials share $\alpha^M$ | 0.462 | 0.756  | 0.959 |
| Labour share $\alpha^L$    | 0.023 | 0.098  | 0.347 |

### 3.6 Sample coverage

Coverage is stable at 33,000–41,000 estimation plants per round (33,199 in 2012 rising to 40,905 in 2021; 39,078 in 2022). Rural plants are 41.8 per cent of the panel in 2012, rising to 45.6 per cent in 2022 (44.0 pooled). The largest divisions by plant-years are

**Table 4:** Sample coverage by round: estimation plant-years and the rural share of the raw factory panel.

| Year | Plant-years | Rural share (%) |
|------|-------------|-----------------|
| 2012 | 33,199      | 41.8            |
| 2013 | 36,396      | 42.6            |
| 2014 | 38,436      | 43.0            |
| 2015 | 37,358      | 43.0            |
| 2016 | 39,099      | 43.8            |
| 2017 | 39,227      | 44.3            |
| 2018 | 39,412      | 44.0            |
| 2019 | 39,750      | 44.8            |
| 2020 | 40,267      | 45.2            |
| 2021 | 40,905      | 45.1            |
| 2022 | 39,078      | 45.6            |

food products (62,535 in the translog sample), non-metallic minerals (39,980), textiles (29,560), chemicals (28,962), machinery (27,019), fabricated metals (22,969) and rubber-plastics (22,780); Appendix Table 13 reports all 23.

## 4 Empirical framework

### 4.1 The markup from cost minimisation

Let plant  $i$  produce  $Y_{it} = F(K_{it}, L_{it}, M_{it}) e^{\omega_{it}}$  and choose the flexible input  $M$  to minimise cost taking its price  $p_{it}^M$  as given, subject to producing  $Y_{it}$ . The Lagrangian  $\mathcal{L} = p_{it}^M M_{it} + \dots + \lambda_{it}(Y_{it} - F(\cdot)e^{\omega_{it}})$  yields the first-order condition  $p_{it}^M = \lambda_{it} \partial Y_{it} / \partial M_{it}$ , where  $\lambda_{it}$  is marginal cost. Multiplying both sides by  $M_{it}/Y_{it}$  and dividing by the output price  $p_{it}$ ,

$$\underbrace{\frac{\partial Y_{it}}{\partial M_{it}} \frac{M_{it}}{Y_{it}}}_{\theta_{it}^M} = \underbrace{\frac{p_{it}}{\lambda_{it}}}_{\mu_{it}} \underbrace{\frac{p_{it}^M M_{it}}{p_{it} Y_{it}}}_{\alpha_{it}^M} \implies \boxed{\mu_{it} = \frac{\theta_{it}^M}{\alpha_{it}^M}} \quad (2)$$

(De Loecker & Warzynski, 2012). Equation (2) deserves unpacking, because everything in the paper flows through it. *What it says:* the output elasticity of materials tells us what fraction of revenue a cost-minimising plant *would* spend on materials if it priced at marginal cost; the observed share tells us what fraction it *does* spend; the ratio of “should” to “does” is exactly the factor by which price exceeds marginal cost. *Why cost-minimisation suffices:* no assumption about demand, market structure, or pricing conduct enters — the plant may be a monopolist, an oligopolist, or a Bertrand competitor; so long as it does not waste money on inputs, (2) holds. This conduct-free property is what lets one formula travel across 23 heterogeneous industries. *What breaks it:* the derivation prices materials at their marginal acquisition cost and lets the plant adjust them freely

within the year. Adjustment costs, long-term contracts, or buyer power over suppliers would each insert a wedge between  $p^M$  and the true shadow price, contaminating the level of  $\mu$  — one reason our emphasis falls on dispersion, trends, and incidence rather than levels, and precisely the failure mode that, applied to labour, becomes our measurement opportunity in Section 4.4. *What remains to be estimated:* the share  $\alpha^M$  is read directly from the accounts; the entire econometric burden falls on the elasticity  $\theta^M$ , to which we now turn.

## 4.2 Technology: translog, and why

With logs  $y, k, l, m$ , the translog gross-output technology is

$$y_{it} = \beta_k k_{it} + \beta_l l_{it} + \beta_m m_{it} + \beta_{kk} k_{it}^2 + \beta_{ll} l_{it}^2 + \beta_{mm} m_{it}^2 + \beta_{kl} k_{it} l_{it} + \beta_{km} k_{it} m_{it} + \beta_{lm} l_{it} m_{it} + \omega_{it} + \varepsilon_{it}, \quad (3)$$

estimated separately for each NIC-2 industry, with plant-specific elasticities

$$\theta_{it}^M = \beta_m + 2\beta_{mm} m_{it} + \beta_{km} k_{it} + \beta_{lm} l_{it}, \quad \theta_{it}^L = \beta_l + 2\beta_{ll} l_{it} + \beta_{kl} k_{it} + \beta_{lm} m_{it}, \quad (4)$$

and  $\theta_{it}^K$  symmetric. The second-order coefficients are what give the translog its economics:  $\beta_{mm}$  lets the materials elasticity vary with materials intensity (technological non-homotheticity), while the cross-terms  $\beta_{km}, \beta_{lm}$  let it vary with the plant’s capital and labour mix — so two plants in the same industry with different input mixes can have genuinely different technologies at their operating points, which is what plant-level heterogeneity in (2) requires. Cobb–Douglas restricts all second-order terms to zero and hence forces every plant in an industry to share one  $\theta^M$ : markup heterogeneity then mechanically equals (inverse) share heterogeneity, assuming away precisely the variation (2) is meant to measure, and loading every share-reporting anomaly straight into the markup. We therefore treat the translog as the headline and Cobb–Douglas (Appendix C) as the diagnostic benchmark: where the two agree (incidence, rankings, trends) we treat the finding as technology-robust; where they disagree (dispersion, levels) the disagreement is itself informative about how much of the measured heterogeneity is technological. Sample means of the translog elasticities are  $\bar{\theta}^M = 0.905$ ,  $\bar{\theta}^L = 0.102$ ,  $\bar{\theta}^K = 0.016$ , with returns to scale  $\approx 1.02$  — the materials-heavy gross-output technology typical of plant data.

## 4.3 Identification: the ACF control function

Estimating (3) by OLS is inconsistent because input choices respond to the productivity the econometrician does not see ( $\mathbb{E}[\omega_{it} | \text{inputs}] \neq 0$ ). The control-function solution proxies  $\omega$  by inverting an observable decision. With materials the proxy (Levinsohn & Petrin, 2003) and the Akerberg et al. (2015) timing — labour chosen with (or before) materials,

capital predetermined — the first stage purges measurement error only:

$$y_{it} = \Phi_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it}, \quad \hat{\Phi}_{it} \text{ estimated by a high-order polynomial}, \quad (5)$$

so that for any candidate  $\beta$ , implied productivity is  $\omega_{it}(\beta) = \hat{\Phi}_{it} - f(k_{it}, l_{it}, m_{it}; \beta)$  with  $f$  the translog aggregate in (3). Productivity is assumed first-order Markov,

$$\omega_{it} = g(\omega_{it-1}) + \xi_{it}, \quad \mathbb{E}[\xi_{it} | \mathcal{I}_{it-1}] = 0, \quad (6)$$

with  $g$  approximated polynomially. The innovation  $\xi_{it}(\beta)$  is then orthogonal to anything in the firm’s  $t - 1$  information set and to predetermined capital, giving the second-stage GMM moments

$$\mathbb{E}[\xi_{it}(\beta) \mathbf{z}_{it}] = 0, \quad \mathbf{z}_{it} = (k_{it}, k_{it}^2, l_{it-1}, l_{it-1}^2, m_{it-1}, m_{it-1}^2, k_{it}l_{it-1}, k_{it}m_{it-1}, l_{it-1}m_{it-1}), \quad (7)$$

current capital and *lagged* flexible inputs instrumenting the translog terms.

The economics behind each piece of (5)–(7) is worth making explicit, since the markup estimates are only as credible as this machinery. *Why OLS fails, and in which direction:* a plant that draws high  $\omega$  buys more materials, so OLS attributes part of productivity to materials and biases  $\theta^M$  upward — and since  $\mu = \theta^M / \alpha^M$ , uncorrected simultaneity would manufacture markups out of productivity. *What the first stage does and does not do:* because materials demand  $m_{it} = m_t(k_{it}, l_{it}, \omega_{it})$  is (under the scalar-unobservable and monotonicity conditions of [Levinsohn & Petrin, 2003](#)) invertible in  $\omega$ , output conditional on  $(k, l, m)$  is purged of everything except the shock  $\varepsilon$ ; but for the same reason no coefficient of (3) is identified in the first stage — conditioning on the inputs that are functions of  $\omega$  absorbs the very variation that would identify them. This is the [Akerberg et al. \(2015\)](#) functional-dependence point, and it is why *all* parameters wait for the second stage. *Why the Markov assumption (6) has content:* it converts “unobserved productivity” from a free object into one with a law of motion, so that its innovation  $\xi_{it}$  is a genuine surprise relative to the plant’s  $t - 1$  information. *Why the instruments in (7) are valid and relevant:* capital is predetermined by time-to-build, so  $k_{it}$  cannot respond to  $\xi_{it}$ ; lagged materials and labour were chosen before  $\xi_{it}$  realised, so they are uncorrelated with it, yet through the persistence of  $\omega$  and of input demands they remain strongly correlated with current input choices — exogeneity from timing, relevance from persistence. *What would break it:* materials mismeasurement that varies with  $\omega$  (violating scalar unobservability), non-Markov productivity (e.g. anticipated demand shifts entering input choices but not  $g$ ), or labour that is fully flexible *and* chosen strictly after materials. The two-industry exclusion and the diagnostic suite in [Appendix E](#) target the first; the robustness of trends and incidence across estimators ([Section 6](#)) speaks to the rest.

We minimise the GMM objective per industry by Nelder–Mead followed by BFGS

(as two optimisation calls), gating the procedure on a single-industry reproduction before the full run; identification requires at least two consecutive observations per plant, which defines the estimation sample. The Cobb–Douglas benchmark uses the same moments via the `prodest` implementation, with analytic standard errors cross-checked against a 50-replication block bootstrap (agreement documented in Section 6).

#### 4.4 The markdown and monopsony

Apply the logic of (2) to labour. If labour were frictionless and competitively priced,  $\theta_{it}^L/\alpha_{it}^L$  would also equal  $\mu_{it}$ . When instead the plant faces an upward-sloping labour supply or wage-setting frictions, the labour condition picks up the additional wedge  $\psi_{it}$  between the marginal revenue product and the wage:

$$\frac{\theta_{it}^L}{\alpha_{it}^L} = \mu_{it} \psi_{it} \quad \Longrightarrow \quad \boxed{\psi_{it} = \frac{\theta_{it}^L/\alpha_{it}^L}{\theta_{it}^M/\alpha_{it}^M}} \quad (8)$$

(Brooks et al., 2021; Yeh et al., 2022). Under static monopsony with firm-level labour-supply elasticity  $\varepsilon_L$ ,

$$\psi = 1 + \frac{1}{\varepsilon_L} \quad \Longleftrightarrow \quad \varepsilon_L = \frac{1}{\psi - 1}, \quad (9)$$

so  $\psi > 1$  diagnoses monopsony and its size maps into the elasticity workers face. The intuition for the boxed ratio in (8) is that the materials condition serves as the plant’s own control group: both input conditions contain the same markup  $\mu$  (both inputs produce the same marked-up output), so dividing the labour ratio by the materials ratio differences the markup out, and whatever wedge survives is specific to the labour transaction. This is why (8) requires no data beyond what (2) already uses — but also why it inherits *both* elasticities’ estimation error, which is why we winsorise  $\psi$  more aggressively and emphasise incidence  $\Pr(\psi > 1)$  over levels. Note also what (8) teaches about measurement: the labour-based “markup”  $\theta^L/\alpha^L$  is the *product*  $\mu\psi$  — markups and markdowns cannot be measured separately from one input, which is why both flexible inputs, and the joint estimation, are essential.

Two caveats on interpretation are owed. First, (9) reads  $\psi$  through the lens of static wage-posting monopsony (Manning, 2003); the measured wedge itself is model-free (it is just  $\text{MRPL}/w$ ), but the elasticity mapping is not — under bargaining, efficiency wages, or dynamic contracts the same  $\psi$  implies different primitives. Second, labour quasi-fixity is an alternative reading: if hiring involves adjustment costs, the static labour condition fails mechanically and  $\psi > 1$  can reflect the shadow cost of adjustment rather than wage-setting power. Three features of our results argue that frictional wage-setting is doing most of the work: the geography ( $\psi$  is highest exactly where outside options are thin — rural plants, mineral-belt states — rather than where turnover or skill-specificity is

highest); the incidence gradient in local labour-market concentration; and the persistence of plant-level  $\psi$  over horizons at which adjustment costs should wash out. We flag the caveat wherever levels, rather than cross-sectional contrasts, carry the argument.

## 4.5 Estimation conventions

Markups are winsorised at the 1st/99th percentile and markdowns at the 2nd/98th within NIC-2×year cells before any aggregation; monetary variables are winsorised 1/99 by year at cleaning. Survey multipliers are never used in estimation (which targets the model) and always used in population aggregation. Monopsony incidence is  $\Pr(\psi_{it} > 1)$ . Two industries are excluded from all markdown aggregates on data-quality grounds established below: printing (NIC 18) and other transport equipment (NIC 30), where 36.4 and 14.9 per cent of plants respectively report  $\alpha^L + \alpha^M > 1$  (negative implied value added, a job-work reporting artefact that destroys (8)’s denominator logic). Concentration is measured by the employment Herfindahl  $HHI_{jt} = \sum_i s_{ijt}^2$  within NIC-4 markets, and aggregate markup dynamics are decomposed by [Melitz & Polanec \(2015\)](#):

$$\Delta \bar{\mu}_t = \Delta \tilde{\mu}_t^{\text{within}} + \Delta \text{cov}_t(s, \mu) + \text{entry}_t + \text{exit}_t, \quad (10)$$

separating within-plant change, reallocation covariance, and net entry margins. The decomposition answers a question the aggregate series cannot: a flat sales-weighted markup is consistent with a competitive economy in stasis, but equally with intensifying reallocation toward high-markup plants masking broad-based price compression — two readings with opposite policy content. Only (10) distinguishes them, and Section 5.2 shows the second reading is the correct one for India.

## 5 Results

### 5.1 Output elasticities

Appendix Table 13 reports, for each of the 23 divisions: sample size, mean  $\theta^L, \theta^K, \theta^M$ , returns to scale, monopsony incidence, and median  $\mu$  and  $\psi$ . Three regularities matter downstream. Materials elasticities cluster tightly (0.82–1.01) — the gross-output technology is materials-dominated everywhere, which stabilises the markup’s numerator. Labour elasticities range 0.05–0.23 with two anomalies (a negative mean in NIC 30 and near-zero monopsony in NIC 18) that motivate the exclusions of Section 4.5. Returns to scale are indistinguishable from constant (0.95–1.08), consistent with the identification assumptions.

## 5.2 Markups: levels, trends, concentration, decomposition

**Table 5:** Translog estimates year by year: median and sales-weighted markup, median mark-down, and monopsony incidence.

| Year | $N$    | med $\mu$ | wtd $\mu$ | med $\psi$ | $\Pr(\psi > 1)$ |
|------|--------|-----------|-----------|------------|-----------------|
| 2012 | 33,130 | 1.248     | 1.272     | 0.870      | 0.450           |
| 2013 | 36,332 | 1.227     | 1.250     | 0.844      | 0.443           |
| 2014 | 38,370 | 1.241     | 1.261     | 0.807      | 0.427           |
| 2015 | 37,322 | 1.272     | 1.290     | 0.722      | 0.391           |
| 2016 | 39,087 | 1.260     | 1.284     | 0.693      | 0.378           |
| 2017 | 39,212 | 1.241     | 1.275     | 0.690      | 0.377           |
| 2018 | 39,402 | 1.225     | 1.252     | 0.708      | 0.386           |
| 2019 | 39,738 | 1.215     | 1.243     | 0.658      | 0.365           |
| 2020 | 40,256 | 1.231     | 1.255     | 0.670      | 0.368           |
| 2021 | 40,899 | 1.197     | 1.208     | 0.745      | 0.403           |
| 2022 | 39,072 | 1.156     | 1.168     | 0.758      | 0.406           |

Table 5 carries the headline series. Reading down the markup columns: the median rises modestly to a 2015 peak (1.272) and falls thereafter to 1.156 in 2022; the sales-weighted series moves in parallel at a small premium. Reading the markdown columns: the median  $\psi$  falls from 0.870 to a 2019 trough of 0.658 and partially rebounds after the pandemic; monopsony incidence traces the same U — 45.0 per cent (2012), 36.5 (2019), 40.6 (2022). The two margins co-move negatively over the decade, a joint dynamic the companion determinants paper links to trade exposure and local labour-market thickening.

**The industry pattern.** The cross-industry ordering (Appendix Table 13) is economically legible: median markups are highest where product differentiation and regulation concentrate — tobacco (1.74), wearing apparel (1.56), pharmaceuticals (1.42), non-metallic minerals (1.38) — and lowest in homogeneous intermediates — basic metals (1.09), textiles (1.19), paper (1.15). Monopsony incidence orders differently: above 60 per cent in food products, petroleum refining, basic metals and coke — process industries whose plants anchor thin local labour markets — and lowest in apparel and textiles, whose mobile, urban workforces face many employers. The rank divergence across the two margins within the same data is itself evidence that the estimation separates them.

The median translog markup by industry spans 1.09 (basic metals) to 1.74 (tobacco), with the plant-weighted aggregate median near 1.22; the economy-wide winsorised mean drifts from roughly 1.32 down toward 1.20 after 2015. Levels of this magnitude are *modest*: they sit at the bottom of the international production-approach range and far below the sales-weighted US aggregates of De Loecker et al. (2020). Three corroborating facts indicate the modesty is real rather than an estimator artefact. First, the Cobb–Douglas benchmark (Appendix C) delivers the same industry ordering and a similar aggregate level

(year-by-year means 1.31–1.38 with no trend), so the level does not ride on the translog. Second, *concentration fell*: the mean NIC-4 employment HHI declined from 0.086 (2012) to 0.066 (2022) and the four-firm concentration ratio from 0.39 to 0.32 — an economy de-concentrating is not one where product-market power is building. Third, the [Melitz & Polanec \(2015\)](#) decomposition (10) of the sales-weighted aggregate (stable at 1.16–1.21 throughout) shows post-2016 *within-plant* markup compression (cumulative  $-0.084$ ) offset by a positive reallocation covariance ( $+0.106$ ): individual plants lost pricing power while activity shifted toward high-markup plants — the same anatomy documented for the US, at one-tenth the level. Entrants price below incumbents in every year, disciplining the margin from below. Table 6 summarises the post-2016 anatomy.

**Table 6:** Melitz–Polanec anatomy of the sales-weighted aggregate markup. The aggregate is flat because within-plant compression and reallocation toward high-markup plants offset.

| Component (cumulative, post-2016)                  | Contribution                     |
|----------------------------------------------------|----------------------------------|
| Within-plant markup change                         | $-0.084$                         |
| Reallocation covariance $\Delta\text{cov}(s, \mu)$ | $+0.106$                         |
| Entrants relative to incumbents                    | below-average $\mu$ , every year |
| Aggregate sales-weighted markup (range, 2012–2022) | 1.16 – 1.21                      |

### 5.3 Markdowns and monopsony

The markdown tells the opposite story. Its median sits below one (0.66–0.87 across years) — the *typical* plant pays labour at or slightly above the static competitive benchmark, consistent with efficiency-wage and regulatory floors — but the distribution’s upper half is heavy: the share of plants with  $\psi > 1$  is 36–45 per cent year by year and 41–44 per cent pooled, and among monopsonistic plants the mean wedge implies, via (9), a labour-supply elasticity near 2 — squarely in the range that modern quasi-experimental estimates recover for much richer labour markets ([Card et al., 2018](#)). Monopsony incidence declines secularly (by five to eight percentage points from 2012 to 2019–20, before a partial post-pandemic rebound), a decade-long erosion of wage-setting power that coincides with the thickening of local labour markets documented in the companion determinants paper.

### 5.4 The aggregation-concept range

Published “aggregate markdowns” differ enormously, and Section G’s algebra shows why: the object is a ratio, and weighting travels through it non-linearly. On a common trimmed sample (400,245 plant-years) we compute three concepts. The *typical-plant* (unweighted) markdown is 1.506 ( $\varepsilon_L \approx 1.98$ ); the *employment-weighted* counterpart 1.119; the *sales-weighted* 4.667 ( $\varepsilon_L \approx 0.27$ ) — large plants mark down hardest per rupee of output; and the [Yeh et al. \(2022\)](#) harmonic aggregate collapses to 0.399, an artefact of harmonic weighting

under extreme dispersion that we document rather than adopt. All concepts agree on incidence (85.1 per cent classification agreement across estimators), on the industry ranking, and on the size gradient; they disagree only on the scalar level. We therefore headline the

**Table 7:** The aggregate markdown under alternative concepts, common trimmed sample ( $N = 400,245$ ). Implied elasticity from  $\varepsilon_L = 1/(\psi - 1)$  where  $\psi > 1$ .

| Concept                         | Estimator    | $\bar{\psi}$ | implied $\varepsilon_L$ |
|---------------------------------|--------------|--------------|-------------------------|
| Typical plant (unweighted mean) | translog     | 1.506        | 1.98                    |
| Employment-weighted mean        | translog     | 1.119        | 8.4                     |
| Sales-weighted mean             | translog     | 4.667        | 0.27                    |
| Harmonic (Yeh et al.)           | translog     | 0.399        | —                       |
| Typical plant (unweighted mean) | Cobb–Douglas | 1.727        | 1.38                    |
| Harmonic (Yeh et al.)           | Cobb–Douglas | 1.305        | 3.28                    |

typical-plant concept and report the range  $\varepsilon_L \in [0.3, 2]$  — treating the sensitivity itself as a finding about the literature.

## 5.5 Size gradients

Within industry-years, the markup rises mildly with size (log-employment coefficient +0.019; +0.048 within plants), consistent with share-based pricing power. The raw markdown-size gradient is contaminated by division bias —  $\alpha^L$ ’s denominator contains output — so we estimate it against *lagged* size: the within-plant markdown gradient on lagged output is +0.075 and on lagged market share +0.033, both precisely estimated. Larger plants both charge more and pay a larger wedge below marginal product — the granular-employer prediction of [Berger et al. \(2022\)](#) in Indian data.

## 5.6 Geography: states and the rural–urban margin

The two margins part ways in space, and the contrast is informative about mechanism. Median markups across major states span only 1.19–1.25: goods travel, so a plant’s pricing power is determined in product markets that are national (or international) in scope, and state boundaries barely register. Monopsony incidence spans 0.09–0.63 across states: workers do not travel nearly as freely, so the labour wedge is set in *local* labour markets and inherits their geography. Incidence is highest in the central-eastern mineral belt (Chhattisgarh  $\approx 0.65$ , Odisha and Jharkhand  $\approx 0.55$ ) — resource-based process industries anchoring one-employer towns — and lowest in the diversified south (Tamil Nadu 0.31), where dense industrial clusters give workers many alternative employers. The theory predicting exactly this pattern is the outside-options logic of wage-posting models ([Manning, 2003](#); [Berger et al., 2022](#)): employer wage-setting power is the inverse of the supply elasticity the plant faces, and that elasticity rises with the density of competing

**Table 8:** Market power across major states (pooled 2012–2022; states with the largest plant counts): median markup, median markdown, monopsony incidence.

| State          | $N$    | med $\mu$ | med $\psi$ | $\Pr(\psi > 1)$ |
|----------------|--------|-----------|------------|-----------------|
| Tamil Nadu     | 52,347 | 1.251     | 0.611      | 0.313           |
| Maharashtra    | 47,501 | 1.230     | 0.633      | 0.339           |
| Gujarat        | 38,814 | 1.213     | 0.838      | 0.438           |
| Uttar Pradesh  | 37,799 | 1.212     | 0.869      | 0.452           |
| Karnataka      | 24,163 | 1.240     | 0.610      | 0.337           |
| Haryana        | 23,363 | 1.235     | 0.693      | 0.374           |
| Rajasthan      | 19,730 | 1.207     | 0.862      | 0.449           |
| West Bengal    | 17,954 | 1.214     | 0.940      | 0.481           |
| Punjab         | 17,935 | 1.223     | 0.844      | 0.443           |
| Andhra Pradesh | 15,039 | 1.200     | 1.038      | 0.514           |
| Madhya Pradesh | 14,560 | 1.194     | 1.092      | 0.529           |
| Kerala         | 12,848 | 1.203     | 0.669      | 0.353           |
| Telangana      | 12,496 | 1.233     | 0.850      | 0.451           |
| Uttarakhand    | 10,509 | 1.245     | 0.994      | 0.498           |

employers. Table 8 reports the major states; Figure 1 maps it. Within states, the sharper margin is sectoral: rural plants are monopsonistic in **44.2 per cent** of plant-years against **38.4 per cent** urban — a gap present in *every* sample year (49.4 vs 44.2 in 2012; 45.3 vs 38.9 in 2022) — while median markups are nearly identical across sectors (1.217 rural, 1.233 urban). Table 9 gives the sector series year by year. Figure 2 maps the gap: it exceeds twenty percentage points in West Bengal (+20.8) and Uttar Pradesh (+20.5), fifteen in Odisha and Telangana, ten in Karnataka, and *reverses* exactly where the thin-market logic predicts — Bihar (−31.7), Assam (−11.7), Punjab (−5.4), states whose rural manufacturing is dominated by large agro-processing employers facing thin urban sectors. Labour-market power in India is a rural, thin-market phenomenon; product-market power has no geography at all.

## 6 Robustness and measurement

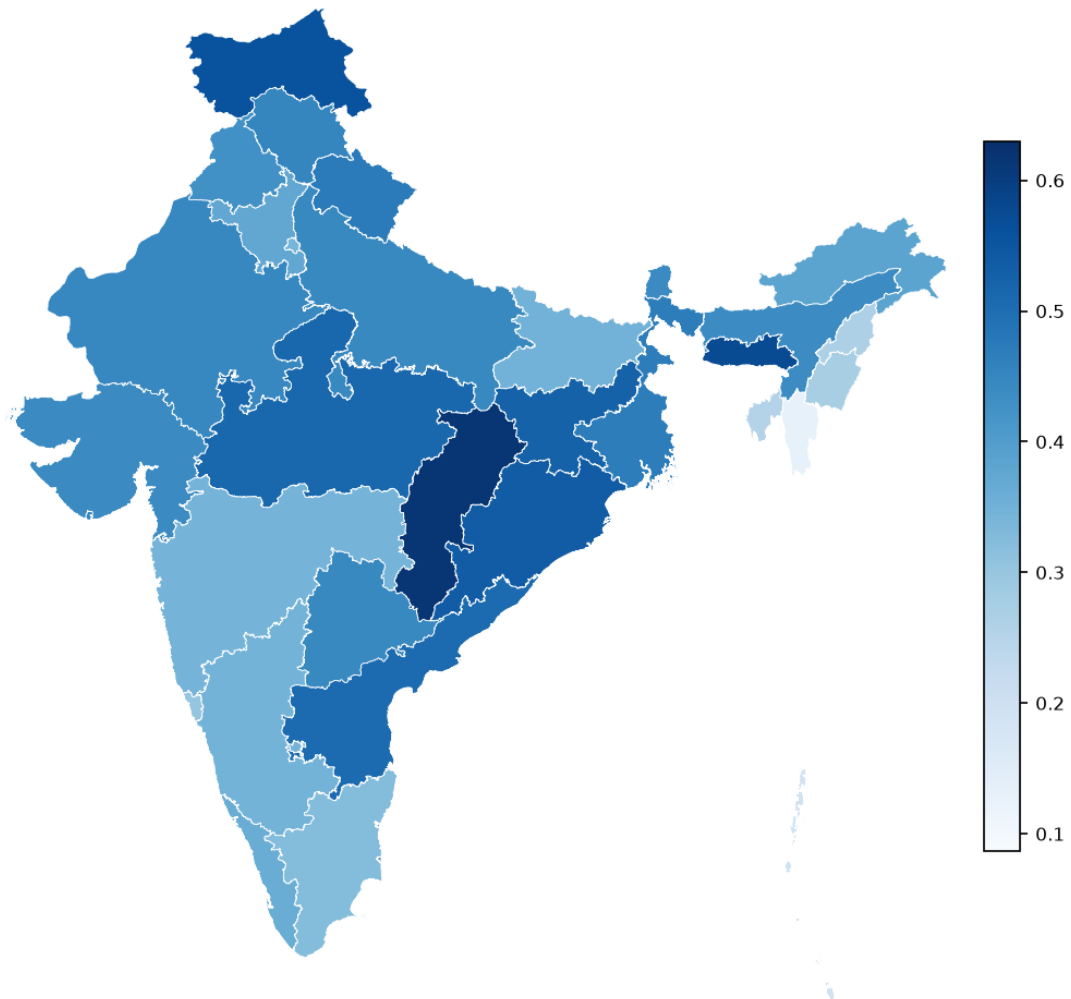
### 6.1 Cross-estimator diagnostics

Translog and Cobb–Douglas agree where they should and disagree where theory says they must. The markdown correlates at Spearman 0.78 across estimators and monopsony classifications agree for 85.1 per cent of plants; the markup rank correlation is lower (0.50) precisely because the translog’s plant-specific  $\theta^M$  absorbs within-industry heterogeneity that Cobb–Douglas forces into the markup. Analytic (Wooldridge-form) standard errors reproduce the bootstrap to within sampling noise (elasticity SEs 0.0015–0.019), and a machine-precision reproduction suite re-derives the NIC-31 estimates exactly.

**Table 9:** The rural–urban margin, year by year: population-weighted mean markup and monopsony incidence by sector. The rural excess in incidence appears in every round.

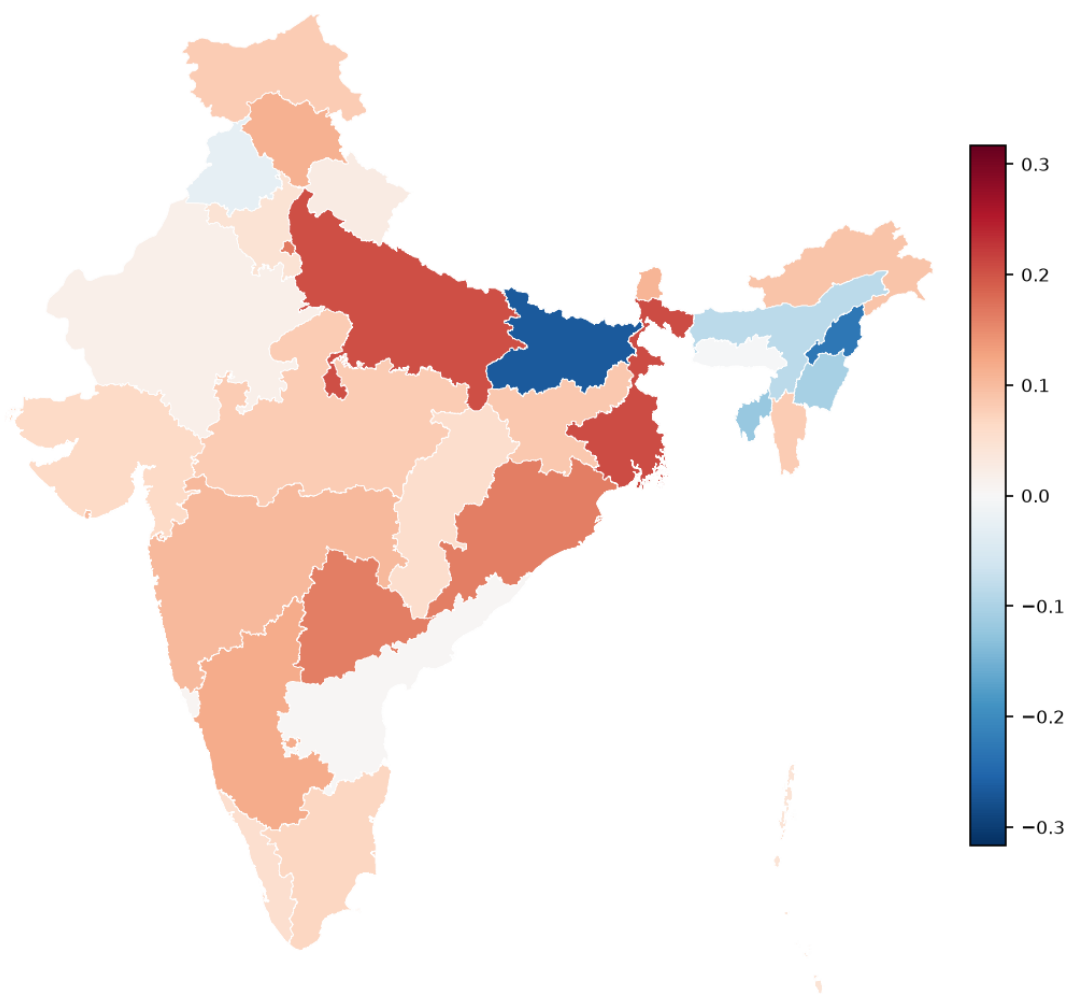
| Year | Weighted mean $\mu$ |       | $\Pr(\psi > 1)$ |       |
|------|---------------------|-------|-----------------|-------|
|      | Rural               | Urban | Rural           | Urban |
| 2012 | 1.295               | 1.256 | 0.494           | 0.442 |
| 2013 | 1.269               | 1.236 | 0.485           | 0.434 |
| 2014 | 1.275               | 1.250 | 0.466           | 0.420 |
| 2015 | 1.297               | 1.285 | 0.429           | 0.384 |
| 2016 | 1.292               | 1.278 | 0.418           | 0.368 |
| 2017 | 1.284               | 1.268 | 0.421           | 0.363 |
| 2018 | 1.261               | 1.244 | 0.431           | 0.370 |
| 2019 | 1.248               | 1.239 | 0.414           | 0.343 |
| 2020 | 1.255               | 1.255 | 0.416           | 0.347 |
| 2021 | 1.217               | 1.202 | 0.452           | 0.383 |
| 2022 | 1.181               | 1.158 | 0.453           | 0.389 |

Monopsony incidence by state (share with  $\psi > 1$ ), 2012-2022



**Figure 1:** Monopsony incidence by state: share of plants with  $\psi > 1$ , pooled 2012–2022. Grey: no/thin data. Source: authors’ estimates from the ASI panel.

Rural minus urban monopsony incidence (pp gap), 2012-2022



**Figure 2:** The rural–urban monopsony gap by state (rural minus urban incidence, percentage points), pooled 2012–2022. Red: rural excess; blue: reversal.

## 6.2 The flagged industries

In NIC 18 (printing) and NIC 30 (other transport), 36.4 and 14.9 per cent of plants report  $\alpha^L + \alpha^M > 1$ : reported factor bills exceed reported output, implying negative value added — a known job-work/contract-manufacturing reporting pattern. Since (8) divides by share-based objects, these cells produce pathological markdowns (NIC 30’s mean  $\theta^L$  is even negative). We exclude both industries from every markdown aggregate; results are insensitive to instead trimming the affected plants.

## 6.3 Bounding the Hicks-neutrality bias

Both estimators impose Hicks-neutral productivity. If technology differences are instead labour-augmenting, measured flexible-input elasticities are inflated and labour elasticities depressed (Demirer, 2025). In the companion methods paper we implement Demirer’s factor-augmenting estimator on the same plants (254,672 matched plant-years) and find: the median markup falls from 1.240 to **1.118** (the revision exceeding two firm-cluster bootstrap standard errors in 18 of 23 industries); the materials elasticity falls 12.3 per cent (almost exactly the bias Demirer reports); and the labour elasticity *rises* by roughly a third. Table 10 summarises the comparison on the matched sample. The implication for the present paper is asymmetric and strengthening: markup *levels* here are upper bounds, while — because  $\psi$  scales with  $\theta^L$  — markdown magnitudes and monopsony incidence are *lower* bounds. The paper’s central contrast survives measurement discipline in the only direction that matters.

**Table 10:** Hicks-neutral versus factor-augmenting estimates on the matched sample ( $N = 254,672$  plant-years; details in the companion methods paper).

|                  | Factor-augmenting | Translog | Cobb–Douglas |
|------------------|-------------------|----------|--------------|
| Median markup    | 1.118             | 1.240    | 1.222        |
| Mean $\theta^M$  | 0.802             | 0.905    | 0.900        |
| Mean $\theta^L$  | 0.159             | 0.102    | 0.112        |
| Mean $\theta^K$  | 0.038             | 0.016    | 0.011        |
| Returns to scale | 0.998             | 1.023    | 1.023        |

## 7 The static welfare cost

Two Harberger-style bounds summarise the stakes. The logic is the textbook triangle: a wedge  $\tau$  between price and marginal cost (or between marginal revenue product and wage) destroys surplus of approximately half the wedge times the distorted expenditure,

the linear approximation to the lost trade. We compute, year by year,

$$\text{DWL}_t^p = \sum_i \frac{1}{2} \left( 1 - \frac{1}{\mu_{it}} \right) s_{it}, \quad \text{DWL}_t^L = \sum_i \frac{1}{2} \left( \frac{\psi_{it}-1}{\psi_{it}} \right) \omega_{it}^w, \quad (11)$$

with  $s_{it}$  the plant’s share of manufacturing sales,  $\omega_{it}^w$  its share of the wage bill (markdown term summed over  $\psi > 1$  plants), and  $1 - 1/\mu$  the plant Lerner index. Three properties of (11) shape its reading. It is *plant-by-plant*, so dispersion matters: the formula is convex in the wedge, and a mean-preserving spread of markups raises the loss — which is why the product-side bound can rise while the average Lerner falls. It is *static and partial*: it counts foregone trades at given technology and plant assignment, not the cross-plant misallocation the same wedges induce, so it is a lower bound on the true cost. And it is *first-order in intent*: we use it to compare margins and track time, not as a welfare level. On the product side, the sales-weighted Lerner index  $1 - 1/\bar{\mu}$  moves from 0.160 (2012) to a 2016 trough of 0.123 and back to 0.143 (2022); the associated deadweight-loss bound (half the wedge times distorted sales, aggregated) drifts from 5.8 to 9.9 per cent of manufacturing sales (peaking at 12.2 per cent in 2019) — rising even as the average Lerner falls, because *dispersion* widened and the triangle formula is convex. On the labour side, the markdown triangle is 2.2–2.8 per cent of the wage bill throughout. Both static

**Table 11:** Static welfare bounds by year: sales-weighted Lerner index; product-market deadweight loss (per cent of sales); labour-market (markdown) deadweight loss (per cent of the wage bill).

| Year | Lerner | DWL <sub>p</sub> (% sales) | DWL <sub>L</sub> (% wage bill) |
|------|--------|----------------------------|--------------------------------|
| 2012 | 0.160  | 5.8                        | 2.6                            |
| 2013 | 0.161  | 7.5                        | 2.5                            |
| 2014 | 0.157  | 7.8                        | 2.3                            |
| 2015 | 0.128  | 7.7                        | 2.3                            |
| 2016 | 0.123  | 9.4                        | 2.2                            |
| 2017 | 0.134  | 8.5                        | 2.3                            |
| 2018 | 0.139  | 8.7                        | 2.4                            |
| 2019 | 0.139  | 12.2                       | 2.3                            |
| 2020 | 0.133  | 8.9                        | 2.3                            |
| 2021 | 0.148  | 7.3                        | 2.6                            |
| 2022 | 0.143  | 9.9                        | 2.8                            |

bounds are dwarfed by the allocative cost the same wedges induce across plants — the subject of the companion misallocation paper, which finds the labour-side wedge is the dominant driver of India’s labour misallocation. Static triangles understate monopsony’s cost precisely because its harm is allocative.

## 8 Conclusion

On the modern Indian factory panel, product-market power is small, stable, spatially uniform, de-concentrating and, under measurement discipline, smaller still; labour-market power is pervasive, geographically concentrated in rural and mineral-belt labour markets, rising with plant size, and understated by standard methods. For policy, the order of priorities follows: the competition question in Indian manufacturing is the price of labour, not the price of goods — employer wage-setting power in thin local markets, invisible to product-market antitrust and to industry-level statistics alike. Three limitations bound the claims. Estimates are revenue-based, so markup *levels* carry the standard caveats — which our factor-augmenting bound quantifies rather than merely acknowledges. The state is the finest geography in this extract, so the rural–urban and state gradients summarise labour markets coarser than commuting zones. And the ASI covers the organised sector only: the wage-setting power facing workers in unorganised manufacturing — most Indian manufacturing employment — is invisible here and plausibly larger. The companion papers take the two natural next steps: tracing what these wedges do to allocative efficiency, and asking what trade policy, exchange rates, and geography do to the wedges.

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## A Panel construction and validation

**Raw structure.** Each ASI round comprises ten fixed-width block files: A (identification: factory id, NIC-2008 activity, state in id characters 6–7, rural/urban item A9, scheme, multiplier), B (organisation), C (fixed assets, by asset class), D (working capital), E (employment and labour cost, by employee category), F (other expenses), G (other output), H (indigenous inputs, by item), I (imported inputs, by item), J (products, by item, with gross sale and ex-factory values). Field positions differ across six format eras (2008–09 through 2022–23); every block of every year is read against a column dictionary parsed from the official layout workbooks, and multi-item blocks are collapsed to the factory’s grand-total row (selector validated against the additive identity of components on the 2012–13 round).

**Table 12:** ASI block structure and its use in this paper.

| Block | Content                  | Use here                                                                          |
|-------|--------------------------|-----------------------------------------------------------------------------------|
| A     | Identification           | factory id; NIC-2008 activity; state (id chars 6–7); rural/urban (A9); multiplier |
| B     | Organisation             | —                                                                                 |
| C     | Fixed assets             | net/gross capital, additions, depreciation (collapsed to totals)                  |
| D     | Working capital          | —                                                                                 |
| E     | Employment & labour cost | persons engaged; wages; employee categories (collapsed)                           |
| F     | Other expenses           | —                                                                                 |
| G     | Other output             | —                                                                                 |
| H     | Indigenous inputs        | materials, by item (collapsed to totals)                                          |
| I     | Imported inputs          | materials, by item (collapsed to totals)                                          |
| J     | Products                 | gross sale and ex-factory output values (collapsed)                               |

**Build chain.** Blocks merged 1:1 on the trimmed 8-character identifier within round; rounds appended: **935,975** factory-years 2008–2022 (identifier×year unique by construction check). Operating filters (positive output, materials, labour, capital; positive multiplier) → **634,124**. Deflation at 2011-12 base (industry WPI for output/materials with documented petroleum and repair fallbacks; machinery WPI for capital; fiscal-year averages of the OEA monthly series) restricts to 2012–2022 → **495,148**. The ACF requirement of a lagged observation within the panel → **423,127** estimation plant-years on **142,493** factories. Validation: per-year medians evolve smoothly across format-era boundaries; the emoluments/wages ratio is 1.10–1.19 throughout; a reproduction suite re-derives calibration and estimation outputs at machine precision.

**Identifier anatomy (verified).** Characters 6–7 of the factory identifier carry the 2011-census state code — stated in the official record layout and confirmed empirically (the frequency ranking reproduces India’s manufacturing geography: Tamil Nadu 12.4%,

Maharashtra 11.2%, Gujarat 9.2%, Uttar Pradesh 8.9% of estimation plant-years). Item A9 (1 = rural, 2 = urban) exists in every round and merges to 100 per cent of estimation observations.

## B Industry-level translog estimates

**Table 13:** Translog DLW estimates by NIC-2008 division, pooled 2012–2022.  $N$  = plant-years;  $\bar{\theta}$  = mean elasticities; RTS = returns to scale;  $\Pr(\psi > 1)$  = monopsony incidence; medians of winsorised  $\mu$  and  $\psi$ . NIC 18 and 30 flagged (Section 4.5): excluded from markdown aggregates.

| NIC2            | $N$    | $\bar{\theta}^L$ | $\bar{\theta}^K$ | $\bar{\theta}^M$ | RTS   | $\Pr(\psi > 1)$ | med $\mu$ | med $\psi$ |
|-----------------|--------|------------------|------------------|------------------|-------|-----------------|-----------|------------|
| 10              | 62,535 | 0.083            | 0.012            | 0.924            | 1.019 | 0.628           | 1.102     | 1.391      |
| 11              | 9,375  | 0.132            | 0.045            | 0.879            | 1.056 | 0.424           | 1.345     | 0.719      |
| 12              | 5,542  | 0.179            | -0.006           | 0.868            | 1.041 | 0.376           | 1.739     | 0.627      |
| 13              | 29,560 | 0.061            | -0.024           | 0.966            | 1.004 | 0.179           | 1.189     | 0.440      |
| 14              | 18,947 | 0.121            | 0.031            | 0.868            | 1.020 | 0.203           | 1.556     | 0.399      |
| 15              | 8,448  | 0.118            | 0.033            | 0.867            | 1.018 | 0.231           | 1.231     | 0.662      |
| 16              | 9,049  | 0.133            | 0.026            | 0.889            | 1.048 | 0.580           | 1.194     | 1.164      |
| 17              | 12,113 | 0.114            | 0.006            | 0.897            | 1.016 | 0.599           | 1.152     | 1.218      |
| 18 <sup>†</sup> | 6,690  | 0.059            | 0.034            | 0.855            | 0.948 | 0.002           | 0.996     | 0.254      |
| 19              | 5,182  | 0.090            | -0.006           | 0.986            | 1.070 | 0.662           | 1.189     | 2.081      |
| 20              | 28,962 | 0.129            | 0.028            | 0.882            | 1.039 | 0.622           | 1.218     | 1.304      |
| 21              | 15,237 | 0.231            | -0.001           | 0.848            | 1.078 | 0.570           | 1.422     | 1.193      |
| 22              | 22,780 | 0.116            | 0.030            | 0.887            | 1.033 | 0.598           | 1.179     | 1.240      |
| 23              | 39,980 | 0.159            | 0.032            | 0.824            | 1.015 | 0.279           | 1.385     | 0.508      |
| 24              | 24,838 | 0.087            | 0.007            | 0.923            | 1.016 | 0.673           | 1.090     | 1.809      |
| 25              | 22,969 | 0.053            | -0.020           | 0.998            | 1.031 | 0.068           | 1.363     | 0.230      |
| 26              | 8,497  | 0.127            | 0.007            | 0.878            | 1.012 | 0.322           | 1.362     | 0.512      |
| 27              | 22,357 | 0.068            | 0.009            | 0.950            | 1.027 | 0.136           | 1.249     | 0.480      |
| 28              | 27,019 | 0.115            | 0.027            | 0.892            | 1.035 | 0.311           | 1.337     | 0.567      |
| 29              | 18,629 | 0.103            | 0.017            | 0.908            | 1.028 | 0.346           | 1.275     | 0.730      |
| 30 <sup>†</sup> | 7,775  | -0.032           | 0.006            | 1.011            | 0.984 | 0.053           | 1.386     | -0.166     |
| 31              | 5,593  | 0.182            | 0.007            | 0.861            | 1.050 | 0.540           | 1.253     | 1.104      |
| 32              | 10,743 | 0.079            | 0.058            | 0.875            | 1.011 | 0.203           | 1.338     | 0.293      |

## C The Cobb–Douglas benchmark

The CD LP/ACF benchmark (one elasticity vector per industry) delivers economy-wide mean markups of 1.31–1.38 across years with no trend, materially identical industry orderings, and the cross-estimator agreement statistics of Section 6. Table 14 reports the by-year series; Table 15 the per-industry estimates.

## D Estimation implementation

**First stage.**  $\Phi_t$  in (5) is a third-order polynomial in  $(k, l, m)$  with year effects, estimated by OLS within industry;  $\hat{\Phi}$  purges only  $\varepsilon$ , so all information about  $\beta$  is deferred to the second stage, per Akerberg et al. (2015).

**Second stage.** For each candidate  $\beta$  (nine translog coefficients),  $\omega_{it}(\beta) = \hat{\Phi}_{it} - f(\cdot; \beta)$ ;  $g$  in (6) is a cubic in  $\omega_{it-1}$ ; the innovation is the residual of that projection. The GMM

**Table 14:** Cobb–Douglas benchmark, by year: mean winsorised markup, labour-based composite  $\theta^L/\alpha^L = \mu\psi$ , and markdown.

| Year | $\bar{\mu}^{CD}$ | $\overline{\mu\psi}^{CD}$ | $\bar{\psi}^{CD}$ | $N$    |
|------|------------------|---------------------------|-------------------|--------|
| 2012 | 1.321            | 2.532                     | 1.911             | 33,199 |
| 2013 | 1.310            | 2.264                     | 1.859             | 36,396 |
| 2014 | 1.318            | 2.211                     | 1.787             | 38,436 |
| 2015 | 1.346            | 2.132                     | 1.679             | 37,358 |
| 2016 | 1.356            | 2.080                     | 1.619             | 39,099 |
| 2017 | 1.353            | 2.128                     | 1.623             | 39,227 |
| 2018 | 1.345            | 2.097                     | 1.673             | 39,412 |
| 2019 | 1.367            | 3.640                     | 1.618             | 39,750 |
| 2020 | 1.376            | 2.415                     | 1.605             | 40,267 |
| 2021 | 1.327            | 2.217                     | 1.805             | 40,905 |
| 2022 | 1.323            | 2.260                     | 1.865             | 39,078 |

**Table 15:** Cobb–Douglas estimates by division: elasticities, returns to scale, markup and markdown.

| NIC2 | $\theta^M$ | $\theta^L$ | $\theta^K$ | RTS   | $\mu^{CD}$ | $\psi^{CD}$ |
|------|------------|------------|------------|-------|------------|-------------|
| 10   | 0.953      | 0.098      | 0.028      | 1.079 | 1.188      | 3.379       |
| 11   | 0.862      | 0.138      | 0.022      | 1.022 | 1.433      | 1.812       |
| 12   | 0.859      | 0.179      | 0.010      | 1.048 | 2.043      | 1.806       |
| 13   | 0.951      | 0.041      | -0.013     | 0.978 | 1.301      | 0.507       |
| 14   | 0.881      | 0.100      | 0.009      | 0.990 | 1.658      | 0.555       |
| 15   | 0.886      | 0.098      | 0.012      | 0.996 | 1.303      | 0.876       |
| 16   | 0.915      | 0.142      | 0.025      | 1.082 | 1.290      | 1.970       |
| 17   | 0.919      | 0.127      | 0.026      | 1.073 | 1.225      | 1.862       |
| 18   | 0.850      | 0.044      | 0.029      | 0.923 | 1.202      | 0.346       |
| 19   | 0.948      | 0.073      | 0.012      | 1.033 | 1.231      | 3.235       |
| 20   | 0.876      | 0.124      | 0.019      | 1.019 | 1.305      | 2.237       |
| 21   | 0.895      | 0.220      | -0.035     | 1.079 | 1.664      | 1.677       |
| 22   | 0.913      | 0.119      | 0.022      | 1.054 | 1.253      | 1.832       |
| 23   | 0.828      | 0.173      | 0.017      | 1.019 | 1.544      | 1.388       |
| 24   | 0.912      | 0.057      | -0.024     | 0.945 | 1.119      | 2.296       |
| 25   | 0.894      | 0.114      | 0.012      | 1.020 | 1.298      | 1.358       |
| 26   | 0.882      | 0.127      | -0.000     | 1.009 | 1.494      | 1.109       |
| 27   | 0.902      | 0.121      | 0.022      | 1.046 | 1.243      | 1.884       |
| 28   | 0.874      | 0.097      | 0.002      | 0.972 | 1.369      | 0.806       |
| 29   | 0.908      | 0.087      | 0.001      | 0.997 | 1.309      | 0.826       |
| 30   | 0.933      | 0.099      | 0.021      | 1.053 | 1.367      | 1.025       |
| 31   | 0.854      | 0.168      | 0.010      | 1.032 | 1.316      | 1.761       |
| 32   | 0.853      | 0.126      | 0.040      | 1.019 | 1.418      | 1.634       |

objective  $Q(\beta) = \left[ \frac{1}{N} \sum \xi_{it}(\beta) \mathbf{z}_{it} \right]' W [\cdot]$  with identity weighting is minimised by Nelder–Mead (restarted) followed by BFGS — two separate optimiser calls, as combining them in one call is not supported in our software — with convergence verified at  $Q \approx 0$  for every industry estimated. Industries with fewer than 500 usable plant-years are not estimated under the translog (only NIC 33, repair, is excluded on this ground). A gate run on a single industry (NIC 31) preceded the full run and is reproduced by the verification suite at machine precision ( $4.5 \times 10^{-17}$  deviation).

**Standard errors.** The Cobb–Douglas benchmark’s analytic (Wooldridge-form) standard errors were cross-checked against a 50-replication plant-block bootstrap; elasticity standard errors of 0.0015–0.019 agree across methods, so analytic errors are reported.

**Order of operations.** Estimation  $\rightarrow$  plant-level  $\mu, \psi \rightarrow$  winsorisation within NIC-2 $\times$ year (1/99 for  $\mu$ , 2/98 for  $\psi$ )  $\rightarrow$  aggregation (with multipliers where population statements are made). Winsorisation *after* estimation and *before* aggregation ensures tails do not drive aggregates while the estimation sample remains the model’s.

## E Data-quality diagnostics

Three diagnostics support the design choices. **(i) Negative implied value added.** On the cleaned manufacturing sample, 32,989 observations report  $\alpha^M > 1$  and 69,031 report  $\alpha^L + \alpha^M > 1$ ; their industry distribution is extremely skewed — 36.4 per cent of plants in NIC 18 (printing) and 14.9 per cent in NIC 30 (other transport) versus low single digits elsewhere — matching the job-work/contract-manufacturing reporting pattern in which the principal’s materials do not transit the contractor’s books. This motivates the two-industry markdown exclusion; markup results are insensitive because materials-share anomalies are winsorised. **(ii) Wage-bill coherence.** The ratio of total emoluments to basic wages lies between 1.10 and 1.19 in every round — the expected loading of bonus, provident fund and welfare over wages — with no format-era discontinuity, validating the era-aware construction of the wage bill. **(iii) Format-era continuity.** Medians of output, materials, employment and capital evolve smoothly across the six raw-format eras, with no breaks at the era boundaries where parsing errors would appear.

## F Division bias in the size gradient, and its resolution

The naive regression of  $\ln \psi_{it}$  on contemporaneous  $\ln L_{it}$  is mechanically biased: from (8),  $\ln \psi = \ln \theta^L - \ln \alpha^L - \ln \mu$  and  $\alpha^L = wL/pY$ , so current  $L$  appears (negatively) inside the dependent variable — classical division bias. An exact log-decomposition on the common estimator sample ( $N = 408,438$ ; identities closing to  $8 \times 10^{-20}$ ) shows the anatomy: the elasticity component  $\ln \theta^M$  contributes *positively* to the size gradient under all three size measures, and the negative raw markup–employment gradient is carried entirely by

the materials-share channel, just as the negative raw markdown–employment gradient is carried entirely by the labour-share channel. The resolution is to measure size pre-determined: on *lagged* size the within-plant markdown gradient is +0.075 (lagged output) and +0.033 (lagged market share), both precisely estimated and cleanly positive — the granular-employer prediction, free of the mechanical channel. We report contemporaneous gradients only alongside this caveat.

## G Aggregation concepts for the markdown

Let plant weights be  $s_i$  (counts, employment, or sales). The *typical-plant* aggregate is  $\bar{\psi} = N^{-1} \sum_i \psi_i$ ; the weighted mean  $\bar{\psi}_s = \sum_i s_i \psi_i$ ; and the [Yeh et al. \(2022\)](#) harmonic aggregate, derived from aggregating plant FOCs,

$$\bar{\psi}^H = \left[ \sum_i \chi_i \psi_i^{-1} \right]^{-1}, \quad \chi_i \text{ output-elasticity-adjusted sales weights.} \quad (12)$$

Because  $x \mapsto 1/x$  is convex, harmonic aggregation loads on the *low- $\psi$*  mass; under the bimodal Indian distribution (a sub-unity mode and a heavy monopsony tail) it collapses below one — 0.399 on our common sample — even though 41–44 per cent of plants mark wages down. The Cobb–Douglas harmonic (1.305) does not collapse, because CD strips the elasticity heterogeneity that interacts with the weights. The implied-elasticity mapping (9) then spans  $\varepsilon_L \in [0.27, 3.3]$  across concept–estimator pairs on identical data: the aggregation concept, not the sample or estimator, is the first-order choice, and papers must state it.